Geometry-dependent identifiability in spectral inverse problems: oblate lifting without prolate improvement

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Abstract

We consider the spectral inverse problem for fluid-loaded elastic shells using the five-mode flexural map $\mathbf{f} = [f_2, \ldots, f_6]^\mathsf{T}$. The equivalent-sphere reduction is rank deficient, while the corresponding five-mode Ritz discretisation produces a positive near-spherical intercept in the smallest singular value that is a numerical artefact rather than a physical effect. Away from the immediate spherical neighbourhood, the oblate branch shows clear growth in the weakest sensitivity direction, whereas a prolate continuation remains less informative. For the canonical parameters, the equivalent-sphere model gives the numerical shadow $\kappa_{\mathrm{eq}} \approx 1.23 \times 10^{10}$, the Ritz sphere gives $\kappa_{\mathrm{Ritz}} = 244$, the canonical oblate point gives $\kappa_{\mathrm{oblate}} = 69.4$, and a representative prolate endpoint at $c/a = 1.5$ gives $\kappa_{\mathrm{prolate}} = 156$. Identifiability therefore depends on how curvature is redistributed, not merely on the loss of spherical symmetry.

1 Introduction

Kac's famous question — can one hear the shape of a drum? — remains a useful way to frame inverse spectral problems, provided one resists the urge to promise more than the data support [1]. In structural acoustics and vibroelasticity, the practical issue is usually local rather than global: given a finite set of measured resonances, can nearby changes in geometry and stiffness be separated reliably? That is a practical-identifiability question in the usual inverse-problem sense [2, 3]. For that purpose, the singular values of a scaled frequency–parameter Jacobian are the natural diagnostic.

The Browntone companion papers established the ingredients needed for this question. Paper 1 introduced the canonical fluid-filled oblate shell used as an abdominal benchmark [4]. Paper 7 showed that an equivalent-sphere reduction can be very effective for a forward problem in which one seeks a dominant stiffness indicator rather than full geometric recovery [5]. Paper 8 then mapped local practical identifiability across oblate and prolate shell families [6]. The present short communication focuses on the corresponding asymmetry result: *oblateness lifts a weak spectral inverse direction, whereas prolateness does not provide a comparable improvement.*

That distinction matters because "symmetry breaking helps" is false in the form in which it is usually repeated. The five-mode flexural map considered here does not reward every departure from spherical geometry. Instead, the improvement is tied to how the perturbed

geometry redistributes curvature across the modal strain field. Oblate shells separate the geometric sensitivity directions enough to improve conditioning; prolate shells, in the same framework, remain less helpful.

Using the canonical fluid-loaded shell and the flexural modes $n = 2, \ldots, 6$ derived from thin-shell theory and fluid–structure coupling [7, 8], we show that the inverse map moves from an equivalent-sphere singular limit to a near-spherical Ritz floor and then to a useful oblate operating point, while the prolate branch remains less well conditioned. The lowest flexural resonance in that map is $f_2 = 3.95\,\mathrm{Hz}$ at the canonical oblate point [4], so the discussion concerns the low-frequency flexural spectrum only; the much higher breathing branch is outside the present scope.

2 Problem Setup

We consider a fluid-filled elastic shell with equatorial semi-axis a, polar semi-axis c, thickness h, Young's modulus E, Poisson ratio ν, wall density ρ_w, fluid density ρ_f, fluid bulk modulus K_f, and internal pressure P_{iap}. The canonical parameter set is that used in the Browntone abdominal-shell studies:

$$a = 0.18\,\mathrm{m}, \quad c = 0.12\,\mathrm{m}, \quad h = 0.01\,\mathrm{m}, \quad E = 0.1\,\mathrm{MPa},$$

$$\nu = 0.45, \quad \rho_w = 1100\,\mathrm{kg/m^3}, \quad \rho_f = 1020\,\mathrm{kg/m^3}, \quad K_f = 2.2\,\mathrm{GPa}, \quad P_{\mathrm{iap}} = 1000\,\mathrm{Pa}.$$

The corresponding aspect ratio is $\zeta = c/a = 0.667$ and the equivalent-sphere radius is $R_{\mathrm{eq}} = (a^2 c)^{1/3} = 0.157\,\mathrm{m}$ [4, 6].

The coupled shell–fluid model is treated by a Rayleigh–Ritz approximation. With Ritz coefficients $\mathbf{q}$, the discrete potential and kinetic energies are

$$\Pi = \tfrac{1}{2}\mathbf{q}^{\mathsf{T}} K(a, c, E)\mathbf{q}, \qquad T = \tfrac{1}{2}\mathbf{q}^{\mathsf{T}} M(a, c)\mathbf{q},$$

where K collects the shell stiffness and fluid-compression contributions, and M collects the shell and added-fluid inertia terms. Stationarity of $\Pi - \omega^2 T$ gives the generalised eigenproblem

$$K(a, c, E)\mathbf{q} = \omega^2 M(a, c)\mathbf{q},$$

from which the resonance frequencies follow as $f_n = \omega_n/(2\pi)$. Papers 1 and 8 give the full derivations of K and M [4, 6], but these equations define the forward model used here.

The inverse variable is the parameter vector $\boldsymbol{\theta} = [a, c, E]^{\mathsf{T}}$. The spectral data vector is formed from the first five flexural resonances,

$$\mathbf{f}(\boldsymbol{\theta}) = [f_2, f_3, f_4, f_5, f_6]^{\mathsf{T}},$$

where f_n is the resonance frequency of flexural mode n. At the canonical oblate point, the lowest member of this set is $f_2 = 3.95\,\mathrm{Hz}$ [4]. We do not use the breathing mode in this inverse map.

Local practical identifiability is measured through the dimensionless scaled Jacobian

$$\widetilde{J} = \mathrm{diag}(\mathbf{f})^{-1} J\, \mathrm{diag}(\boldsymbol{\theta}), \qquad \widetilde{J}_{ij} = \frac{\theta_j}{f_i}\frac{\partial f_i}{\partial \theta_j}, \qquad J_{ij} = \frac{\partial f_i}{\partial \theta_j},$$

with singular values $\sigma_1 \geq \sigma_2 \geq \sigma_3 \geq 0$. The conditioning metric is

$$\kappa = \frac{\sigma_1}{\sigma_3},$$

provided $\sigma_3 > 0$. Large κ indicates a weakly observable parameter combination; $\sigma_3 = 0$ implies local rank deficiency.

Three geometry classes are compared.

1. **Equivalent sphere.** The forward model depends on a and c only through R_{eq}. The geometry columns of the exact Jacobian are therefore linearly dependent, so the map is singular in the strict algebraic sense [5, 6].

2. **Oblate Ritz family.** The full spheroidal curvature field is retained. For the oblate branch we use the eccentricity $\varepsilon = \sqrt{1 - c^2/a^2}$ and the aspect-ratio offset $\delta = 1 - c/a$, and examine the near-spherical limit together with the canonical oblate point.

3. **Prolate Ritz family.** The same five-mode diagnostic is applied to a prolate continuation away from the sphere, so that the effect of geometry class can be separated from the mere fact of leaving SO(3) symmetry behind. Along this branch $c > a$, the enclosed volume and wall thickness are fixed, and the path is parameterised by $c/a \in [1.0, 1.5]$ with a decreasing and c increasing subject to $a^2 c = \mathrm{const} = R_{\mathrm{eq}}^3$.

3 Results

Table 1 reports the singular values of the scaled Jacobian for the five-mode map $\mathbf{f} = [f_2, \ldots, f_6]^\mathsf{T}$, using central differences with relative step $\Delta\theta_j/\theta_j = 1 \times 10^{-4}$. The equivalent-sphere row is the baseline: because that model depends on a and c only through R_{eq}, the exact Jacobian is rank deficient and the reported $\sigma_3 = 1.54 \times 10^{-10}$ is only the floating-point residue of a zero singular value. The Ritz sphere is therefore a distinct calculation. It retains the same parameter vector and mode set, but evaluates the full spheroidal Ritz model at $a = c$.

The oblate branch then shows the relevant lifting behaviour. Moving from the Ritz sphere to the near-oblate point at $\delta = 1 - c/a = 0.1$ raises the weakest singular value from $\sigma_3 = 0.008{,}06$ to $\sigma_3 = 0.012{,}74$, and the canonical oblate point ($c/a = 2/3,\ \delta \approx 0.333$) raises it further to $\sigma_3 = 0.026{,}43$. The corresponding condition number falls from $\kappa_{\mathrm{Ritz}} = 244$ to $\kappa = 145$ and then to $\kappa_{\mathrm{oblate}} = 69.4$. Since the inverse variables and measured mode set are unchanged, that improvement is geometric rather than procedural.

The prolate continuation remains less informative. Along this branch the volume and wall thickness are fixed, with a decreasing and c increasing so that $a^2 c = \mathrm{const}$. At the representative endpoint $c/a = 1.5$ the scaled Jacobian gives $\sigma_3 = 0.016{,}36$ and $\kappa = 156$, which is better than the Ritz sphere but still markedly worse than the canonical oblate configuration.

For the five-mode oblate map parameterised by $\delta = 1 - c/a$, the near-sphere fit

$$\sigma_{\min}(\delta) = \sigma_0 + \lambda_1 \delta^2$$

Table 1: Singular values of the scaled Jacobian $\widetilde{J} = \mathrm{diag}(\mathbf{f})^{-1} J\, \mathrm{diag}(\boldsymbol{\theta})$ for the five-mode inverse map $\mathbf{f} = [f_2, \ldots, f_6]^{\mathsf{T}}$. Central differences use $\Delta\theta_j/\theta_j = 1 \times 10^{-4}$.

Case	c/a	δ	σ_1	σ_2	σ_3	κ
Equivalent sphere	1.0	0	1.893	0.112	1.54×10^{-10}	1.23×10^{10}
Ritz sphere	1.0	0	1.971	0.573	0.008,06	244
Near oblate	0.9	0.1	1.854	0.436	0.012,74	145
Canonical oblate	2/3	0.333	1.835	0.130	0.026,43	69.4
Prolate endpoint	1.5	—	2.551	0.382	0.016,36	156

gives $\sigma_0 = 0.0107$ with 95% confidence interval $[0.0098, 0.0117]$ and $\lambda_1 = 0.137$ with 95% confidence interval $[0.120, 0.155]$. This compact fit summarises the oblate trend in Table 1; its intercept is an extrapolated parameter of the quadratic fit, not a replacement for the directly computed Ritz-sphere value $\sigma_3 = 0.008{,}06$ listed in the table.

The positive intercept at the Ritz sphere is not interpreted as a physical effect. At $a = c$ the underlying continuous problem is symmetric under $a \leftrightarrow c$, but the finite Ritz basis breaks that symmetry through the spheroidal integration metric and the near-sphere fluid blending used in the added-fluid terms. The intercept also grows rather than converges away in the tested truncations, with $\sigma_0 \approx 0.006$, 0.011, and 0.018 for the three-, five-, and seven-mode models, respectively, which is the opposite of convergence. We therefore treat σ_0 as a discretisation artefact and not as a genuine spherical lifting mechanism.

The scientific conclusion rests on the deformation-dependent growth away from the immediate spherical neighbourhood. Once the shape moves into the regime in which the quadratic shape term governs the change in $\sigma_{\min}$—for the present branch, outside the immediate near-sphere interval and in practice for oblate deformations with $\varepsilon > 0.2$, equivalently $\delta \gtrsim 0.02$—the oblate branch separates the weak inverse direction more effectively than either the equivalent-sphere model or the representative prolate continuation. In that sense the reported oblate floor is conservative: the artificial positive intercept raises the near-spherical baseline and therefore understates the gain obtained at the canonical oblate point.

The practical meaning is straightforward. The second flexural mode $f_2 = 3.95\,\mathrm{Hz}$ already lies in the low-frequency band used by the companion studies, and the higher modes in the set respond differently to the same curvature perturbation. That differential response separates the sensitivity directions for a, c, and E in the oblate family. The prolate family does not deliver the same separation, so loss of spherical symmetry is not, by itself, a useful identifiability category.

4 Discussion

The main lesson is geometric and local. We are not proving that spectral data globally determine shell shape, nor even that every non-spherical perturbation helps. We are showing that, for the present five-mode inverse map, the recoverability of (a, c, E) depends strongly on *which* geometry family is used to leave the spherical limit. The difference between $\kappa_{\mathrm{oblate}} = 69.4$ and $\kappa_{\mathrm{prolate}} = 156$ is too large to dismiss as a numerical quirk.

The physical explanation is mode-dependent curvature sampling. In the Ritz model, different flexural modes weight the equatorial and polar regions differently in the strain-energy integrals. Oblate deformation redistributes curvature in a way that makes the a- and c-sensitivity columns less nearly parallel, so the smallest singular value grows. The equivalent-sphere reduction suppresses that mechanism completely because both geometric variables are routed through R_{eq} alone. The prolate family preserves the formal break from spherical symmetry but does not generate an equally useful separation of sensitivity directions.

This asymmetry also clarifies the relation between the companion papers. Paper 1 motivates the abdominal shell as an oblate, fluid-loaded structure whose low-order flexural modes sit in the infrasonic band [4]. Paper 7 shows why an equivalent-sphere model can still be serviceable when the engineering question is dominated by one effective stiffness scale [5]. Paper 8 demonstrates that the moment one asks for joint recovery of geometry and material parameters, that reduction can become the problem rather than the cure [6]. The present calculations show that this loss of information is relieved more effectively by oblate than by prolate curvature redistribution.

Several limits should be kept in view. The analysis is local, based on the Jacobian at a chosen operating point. It uses five low-order flexural modes, not a complete spectrum. It is tied to the present shell parameterisation and to the fluid-loaded Ritz formulation grounded in classical shell and fluid–structure theory [7, 8]. None of this implies that prolate bodies are unrecoverable under all measurements, only that the current spectral map does not improve appreciably when the geometry is made prolate. If such cases must be inverted, additional observables or external priors will be required.

5 Conclusion

Within the five-mode spectral inverse problem for a fluid-filled shell, the equivalent-sphere reduction is singular in exact arithmetic and appears numerically as $\kappa_{\mathrm{eq}} \approx 1.23 \times 10^{10}$. The corresponding five-mode Ritz discretisation at $a = c$ gives a finite $\kappa_{\mathrm{Ritz}} = 244$, but the associated positive intercept in $\sigma_{\min}$ is a discretisation artefact, not a physical spherical lifting. Away from the immediate spherical neighbourhood, the oblate branch increases $\sigma_{\min}$ and reaches $\kappa_{\mathrm{oblate}} = 69.4$ at the canonical geometry, whereas a representative prolate endpoint at $c/a = 1.5$ gives $\kappa_{\mathrm{prolate}} = 156$.

The conclusion is therefore specific rather than universal: identifiability in this spectral inverse problem is geometry-dependent. Oblate curvature redistribution lifts the weak inverse direction; prolate distortion does not deliver the same improvement.

References

[1] Mark Kac. Can one hear the shape of a drum? *The American Mathematical Monthly*, 73(4):1–23, 1966.

[2] Andreas Raue, Clemens Kreutz, Thomas Maiwald, Julie Bachmann, Marcel Schilling, Ursula Klingmüller, and Jens Timmer. Structural and practical identifiability analysis of

partially observed dynamical models by exploiting the profile likelihood. *Bioinformatics*, 25(15):1923–1929, 2009.

[3] Franz-Georg Wieland, Adrian L. Hauber, Marcus Rosenblatt, Christian Tönsing, and Jens Timmer. On structural and practical identifiability. *Current Opinion in Systems Biology*, 25:60–69, 2021.

[4] Jonathan Mace and Brian R. Mace. Modal analysis of a fluid-filled viscoelastic oblate spheroidal shell: Implications for the existence of the "brown note". Companion paper (Paper 1), Journal of Sound and Vibration manuscript, 2026.

[5] Jonathan Mace and Brian R. Mace. Can you hear the ripeness? an equivalent-sphere shell model for acoustic estimation of rind stiffness in watermelons. Companion paper (Paper 7), manuscript, 2026.

[6] Jonathan Mace and Brian R. Mace. Can you hear the shape of an organ? practical identifiability of elastic shell parameters from resonant frequencies. Companion paper (Paper 8), manuscript, 2026.

[7] Horace Lamb. On the vibrations of an elastic sphere. *Proceedings of the London Mathematical Society*, s1-13:189–212, 1882.

[8] Miguel C. Junger and David Feit. *Sound, Structures, and Their Interaction*. MIT Press, Cambridge, MA, 2nd edition, 1986.